Vector → Scout

Tier 1 Modeling Background, Direction, and Immediate Coding Objectives

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Purpose of This Document

Scout,

This document is intended to synchronize you with the current conceptual and modeling direction of the broader research effort.

Charles and Vector have been refining: - the theoretical framing, - the variable ontology, - and the early-stage modeling architecture

behind the basketball findings that successfully replicated the Army results.

The goal now is to transition from: - empirical discovery, toward: - interpretable mechanism modeling and validation.

This document summarizes: 1. where the project currently stands, 2. the emerging modeling architecture, 3. the phased ("Tiered") modeling strategy, 4. immediate coding objectives, 5. and recommended documents for review.

Big Picture

The emerging theory concerns:

$$\textbf{advancement under constrained distinction}$$

The core idea is that: - individuals compete inside local talent pools, - while advancement decisions occur globally under finite selection capacity.

Examples: - Army officers competing locally for Top Blocks while promotion is global, - NCAA players competing locally for minutes/roles while NBA draft selection is global, - academics competing locally for recognition while tenure systems allocate finite distinction.

The basketball domain currently provides the cleanest environment for early formal modeling because:

$$\textbf{selection pool is global} \quad \text{while} \quad \textbf{opportunity pool is local}$$

NBA draft selection is global.

However: - minutes, - role, - touches, - visibility, - and usage

are allocated locally within teams.

This local/global distinction is now one of the central conceptual foundations of the framework.

Current Empirical Finding

The replicated empirical result is an inverted-U relationship between: - local teammate quality, and: - probability of eventual NBA draft selection.

Current operationalization: - OPM (own performance metric) = points per minute (PPM), - local pool quality = leave-self-out teammate PPM, - success event = eventual NBA draft selection.

Empirically: - stronger teammate environments initially improve advancement probability, - but at sufficiently high local talent concentration, the effect reverses.

This suggests: - developmental benefits, - and competitive congestion

may coexist simultaneously.

Current Modeling Direction

The current direction is intentionally:

$$\textbf{minimalist and mechanistic}$$

The immediate goal is NOT: - full realism, - causal identification, - or complete system simulation.

Instead, the goal is:

$$\textbf{identify the minimal mechanism set capable of reproducing the observed nonlinear shape.}$$

This distinction is important.

The working philosophy increasingly resembles: - Santa Fe Institute style modeling, - Dashun Wang style mechanism discovery, - and complex systems minimalism.

Tiered Modeling Strategy

The project is currently organized conceptually into "tiers."

Tier 1 — Minimal Structural Models

Current focus.

Question:

$$\textbf{Can finite opportunity and constrained distinction alone generate the observed nonlinear relationship?}$$

Tier 1 intentionally excludes: - complex network effects, - endogenous sorting, - dynamic adaptation, - and advanced strategic behavior.

The current Tier 1 ingredients are:

$$\textbf{heterogeneous local pools} + \textbf{finite global selection capacity} + \textbf{local opportunity competition} \rightarrow \textbf{nonli}$$

Current modeling document:

> `tier_1_model_v4.md`

This document now contains: - notation, - ontology, - candidate congestion measures, - variable definitions, - glossary, - and preliminary formulations.

Tier 2 — Network Exposure Models

Future direction.

Tier 2 introduces: - nearest-neighbor exposure, - hierarchical networks, - talent centers of gravity, - and network topology.

Examples: - teammate exposure networks, - Army hierarchical structures, - prestige pipelines, - assignment networks.

Important notation clarification:

Nearest-neighbor notation:

$$(\cdot)_{nn}$$

should ONLY be used relative to a specified graph.

For example: - teammate network, - hierarchy network, - rating network, - developmental network, etc.

This distinction is theoretically important.

Tier 3 — Strategic Sorting / Dynamic Systems

Longer-term direction.

Possible future ingredients: - endogenous environment selection, - prestige vs congestion tradeoffs, - strategic positioning, - equilibrium clustering, - self-organizing talent distributions.

NOT current implementation priority.

Current Tier 1 Variables

The current framework distinguishes between:

Variable Type	Example
latent variables	A_i
empirical proxies	PPM
constructed local statistics	$Q_{jt}^{(-i)}$
congestion measures	C_{ijt}
outcomes	Y_i

This distinction is important for implementation clarity.

Current Candidate Congestion / Pressure Variables

One of the biggest recent conceptual developments is distinguishing:

$$Q_{jt}^{(-i)} \neq C_{ijt}$$

where: - $Q_{jt}^{(-i)}$ = average local pool quality, - C_{ijt} = local congestion / roster pressure.

This is theoretically important because: - average teammate quality, and: - competition for scarce opportunity

may represent different mechanisms.

Candidate 1 — Threshold Pressure

$$C_{ijt}^{\tau} = \sum_{\ell \in P_{jt}, \ell \neq i} \mathbf{1}(A_{\ell jt} > \tau)$$

Interpretation: - number of teammates above a meaningful performance threshold.

Candidate 2 — Total Roster Pressure

$$C_{ijt}^{sum} = \sum_{\ell \in P_{jt}, \ell \neq i} A_{\ell jt}$$

Interpretation: - cumulative local talent pressure generated by surrounding teammates.

Current expectation: different congestion formulations may behave differently across domains.

For now: treat congestion as a family of candidate variables.

Immediate Coding Objectives

The immediate mission is NOT: - publication-ready causal inference, - or perfect specification.

The current objective is:

$$\textbf{rapid exploratory mechanism validation}$$

Specifically:

Phase 1 — Variable Construction

For every player-season:

Construct: - player PPM, - leave-self-out teammate PPM, - candidate congestion variables, - opportunity measures, - eventual draft outcome.

Potential variables:

Variable	Meaning
A_i	player ability proxy (PPM)
$Q_{jt}^{(-i)}$	teammate quality
C_{ijt}^{sum}	roster pressure
C_{ijt}^{τ}	threshold congestion
O_{ijt}	minutes / opportunity
Y_i	eventual draft outcome

Phase 2 — Exploratory Modeling

Potential next steps: - logistic models, - nonlinear terms, - interaction terms, - visualization, - response surfaces, - partial dependence, - congestion decomposition.

Goal: determine whether congestion and opportunity variables help explain: - the observed inverted-U, - or its turning point.

Phase 3 — Toy Simulation Worlds

Future near-term objective.

Construct simplified simulated environments containing: - latent ability, - local pools, - finite opportunity, - global selection.

Question:

$$\textbf{Can the inverted-U emerge from these ingredients alone?}$$

This is scientifically important because: - if the minimal structure reproduces the empirical shape, then: - the theory gains mechanistic credibility.

Recommended Documents for Scout to Review

Highest Priority: - `tier_1_model_v4.md`

Strongly Recommended: - `Vector_Master_Theory_and_Modeling_Notes.md` - `2026_0430_Paper7_feedback.md` - `Vector_Questions_and_Modeling.md` - `Vector_HS_Data_Pathway.md` - `Vector_to_Scout_HS_data_thoughts_etc.md`

If available: - Army conceptual background notes, - Basketball replication plots, - Any latest empirical notebooks/pipelines.

Important Conceptual Clarifications

1. OPM

"OPM" refers generally to: - own performance metric.

Examples: - basketball → PPM, - Army → TB ratio / evaluation quality, - academia → publications/citations.

In the current formal notation: - OPM often operationalizes latent ability A_i.

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2. Local vs Global

This distinction is foundational.

Basketball: - local opportunity allocation, - global advancement selection.

Army: - local evaluation, - global promotion boards.

Academia: - local departmental evaluation, - broader prestige and tenure markets.

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3. Nearest Neighbor Notation

Do NOT casually use: - “peer average,” - “nearest neighbor,” - and “pool average”

interchangeably.

Nearest-neighbor notation:

$$(\cdot)_{nn}$$

must always be defined relative to: - a specific graph/network structure.

Current Strategic Direction

The project is increasingly transitioning from:

> *“interesting empirical finding”*

toward:

> *“candidate general theory of advancement under constrained distinction.”*

The current strategic emphasis is: - conceptual clarity, - variable discipline, - minimal models, - interpretable mechanisms, - and iterative empirical validation.

References / Influences

Important conceptual influences include:

- Dashun Wang et al. (2019), *Quantifying the dynamics of failure across science, startups and security*
- Evans-related work on finite local pools and structural misclassification
- Friendship paradox / nearest-neighbor exposure literature
- Network science exposure models

- Complex systems minimal-mechanism modeling traditions

Final Note

Current priority: - get the Tier 1 variables operationalized cleanly, - test candidate congestion formulations, - and begin exploratory model fitting/simulation.

The immediate goal is not perfection.

The immediate goal is:

$$\textbf{determine which minimal structural ingredients appear necessary to reproduce the empirical phenome}$$